

FPP Remote pHAT for Raspberry Pi Zero 2 W: 4-port RS-422 (Falcon differential) pixel driver, rev D (65 x 30 mm, same outline as the Pi Zero)

Pi mounts under this board on 11 mm standoffs (M2.5 holes H1-H4, 58 x 23 mm). Power: 12 V in on J4 -> F1 -> D1 -> U4 -> 5 V onto Pi pins 2/4.
NEVER power the Pi from its USB while 12 V is connected (a Pi Zero has no diode between its USB 5 V input and the header 5 V pins).
FPP: cape EEPROM U3 on i2c-1 @0x50 (fppcapedetect registers a 24c256). DPIPixels strings: port1=P1-29, port2=P1-31, port3=P1-26, port4=P1-7.
RJ45: port1 = pins 1(+),2(-) port2 = 3(+),6(-) port3 = 4(+),5(-) port4 = 7(+),8(-). Y = +, Z = -. Driver enables G and -G tied low.
JP1 open (default): EEPROM writable so FPP can program and sign it in place. Close JP1 afterwards to write-protect (WP high).

